

NPTEL Course on "Computer System Optimization for ML"

Lecture #33

NPTEL

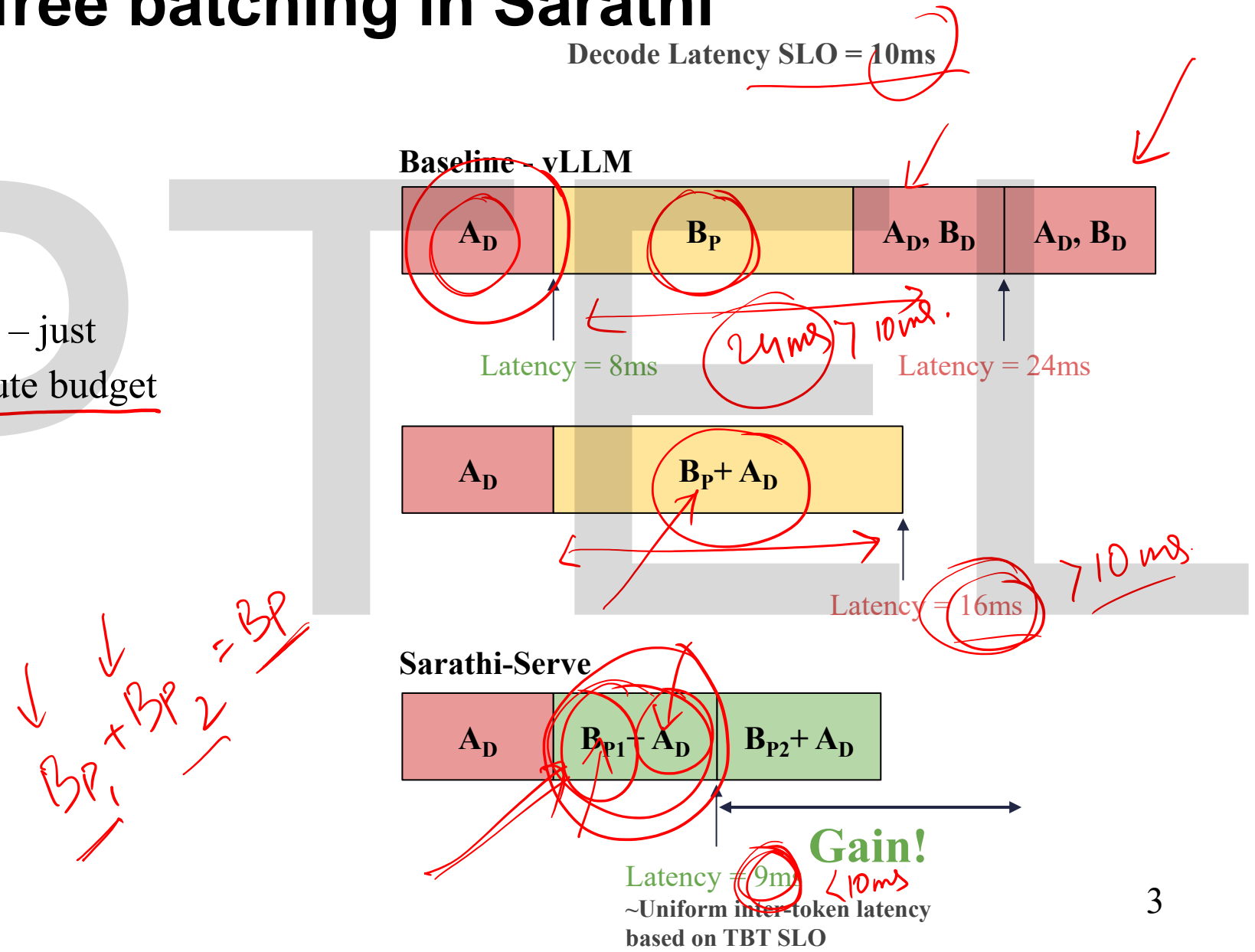
Agenda

- Sarathi-serve (contd.)
- Distributed Service

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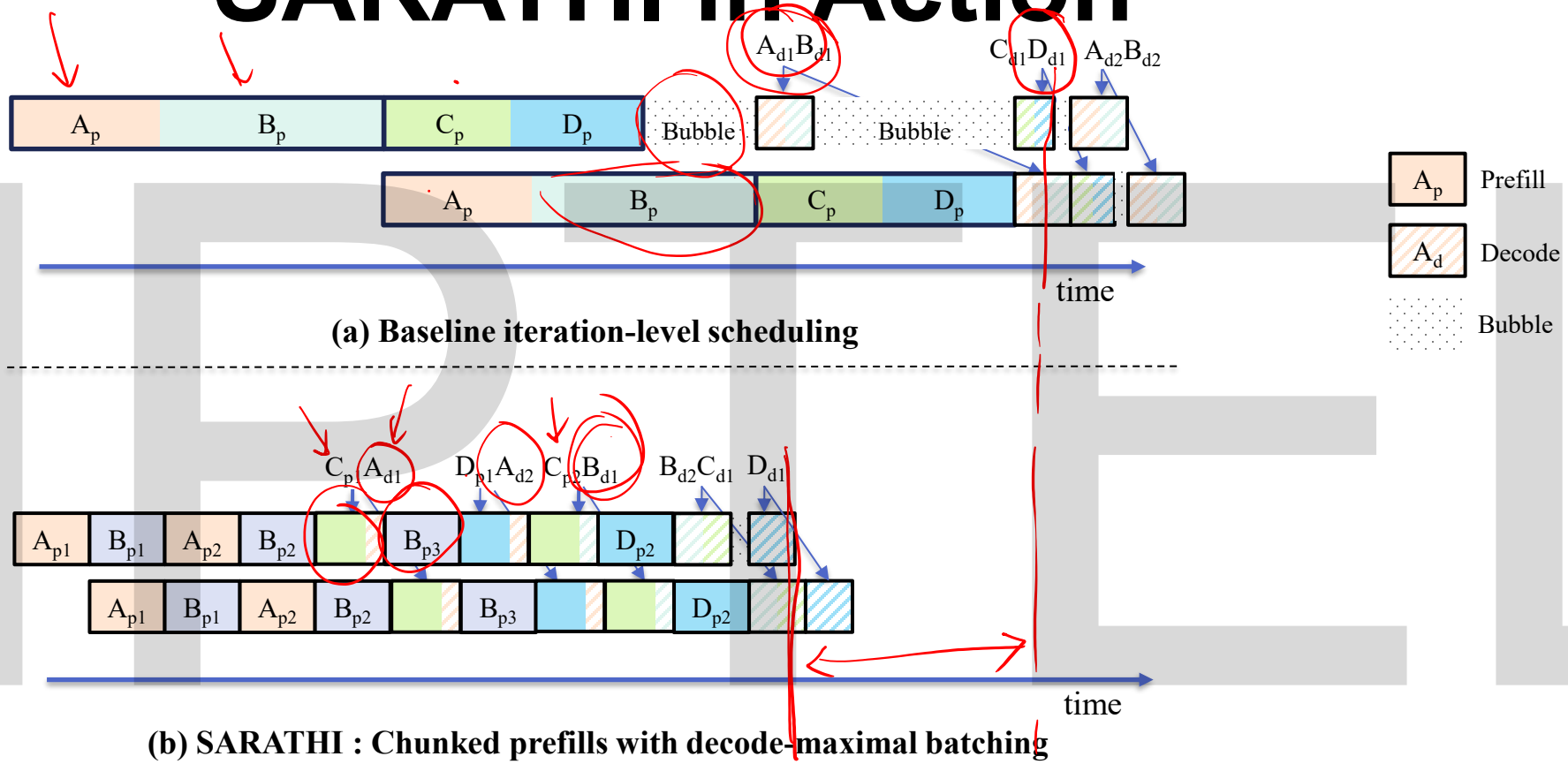
Stall-free batching in Sarathi

Split large prefills into smaller chunks – just enough to consume the leftover compute budget in decode batches



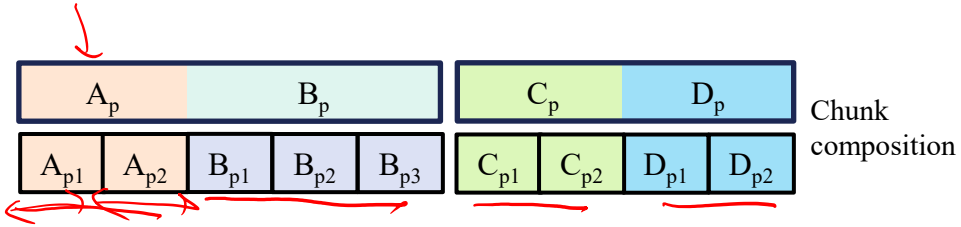
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SARATHI in Action



(a) Baseline iteration-level scheduling

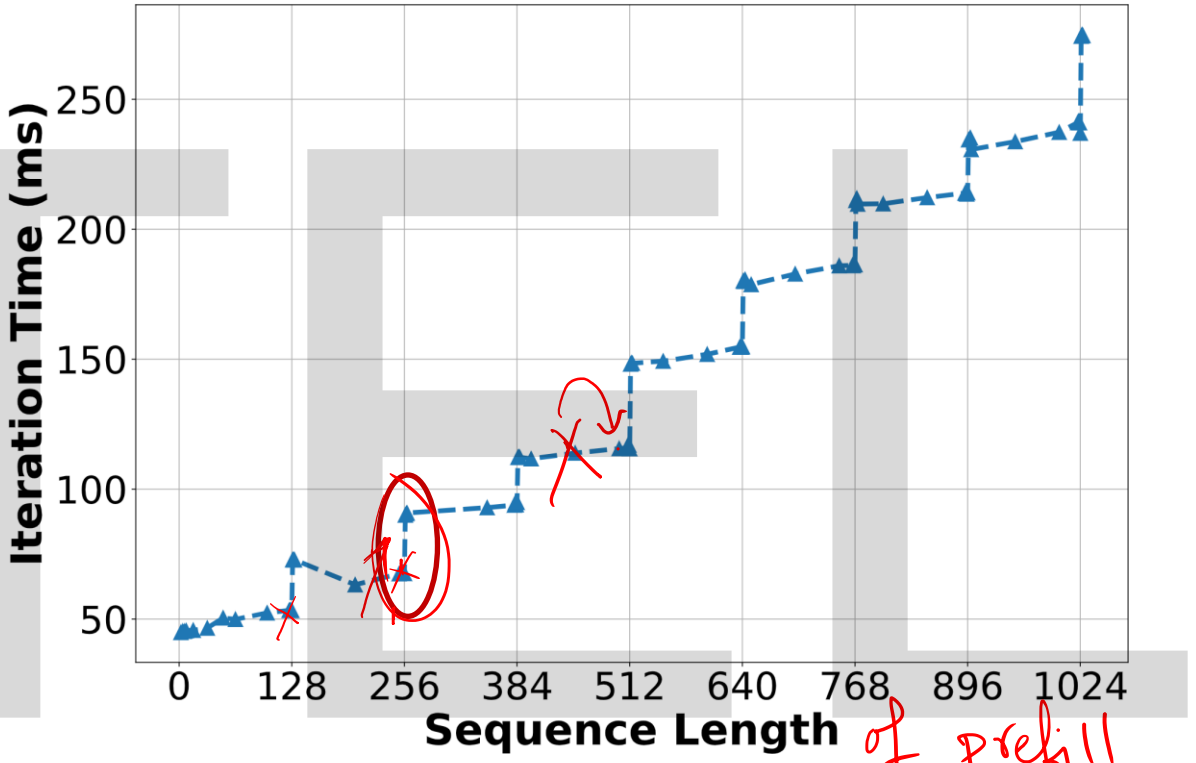
(b) SARATHI : Chunked prefills with decode-maximal batching



Picking Chunk Size

- **Chunk that meets TBT**
- **Larger chunk is not always better**
- **"Tile-quantization" in GPU matmuls**
  - Extraneous computation if tensor shapes not a multiple of the tile size
- **Example (per token cost):**

Chunk size	Per token time (us)
128	0.43
256	0.27
257	0.36



Each chunk should be a multiple of the tile size

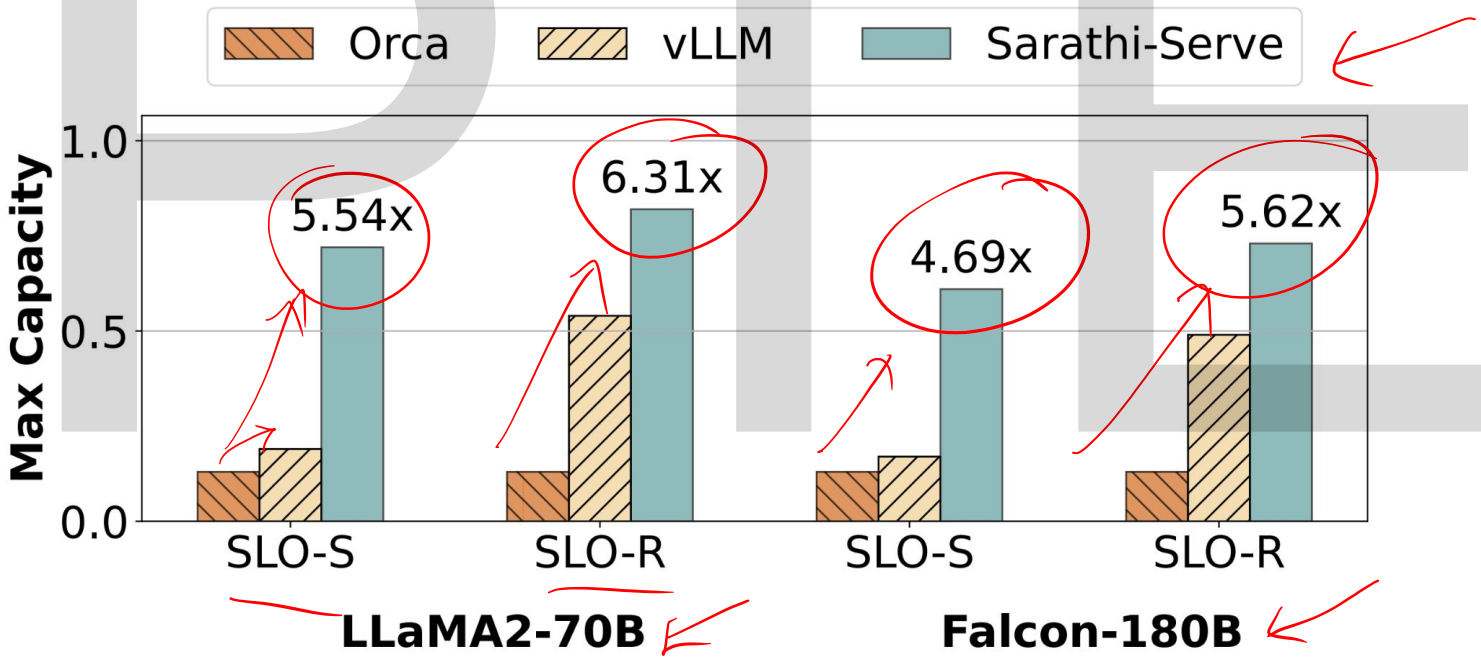
NOT FINAL

Serving Capacity under SLOs

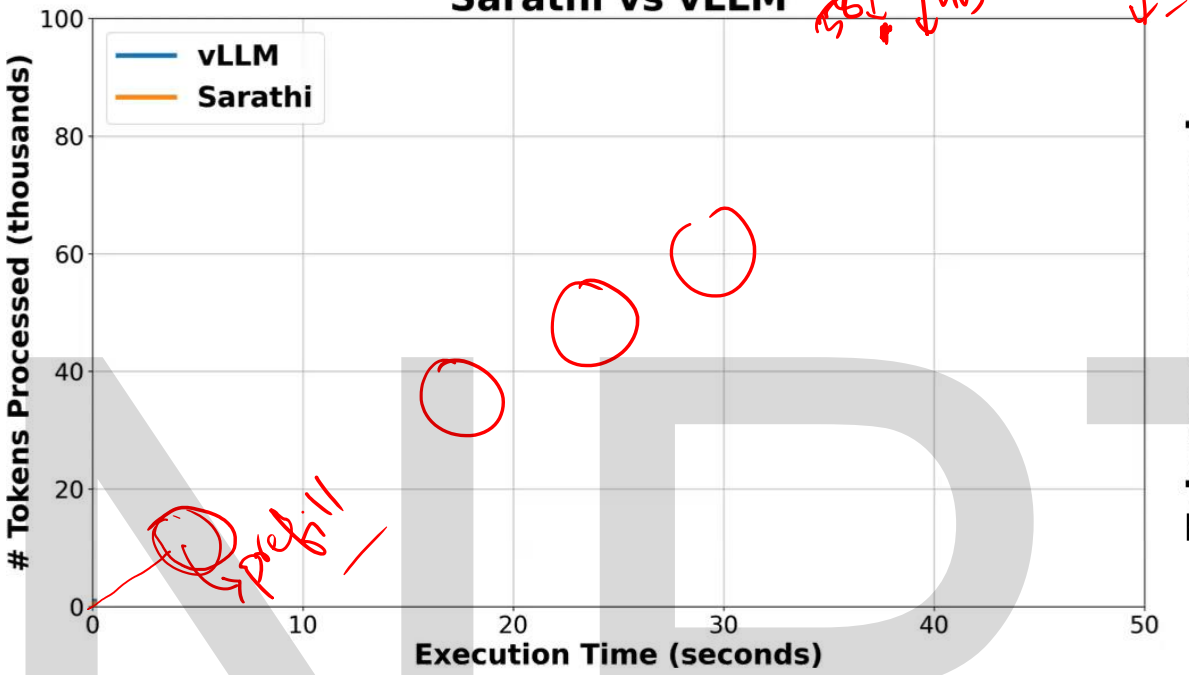
adapt using different chunk sizes

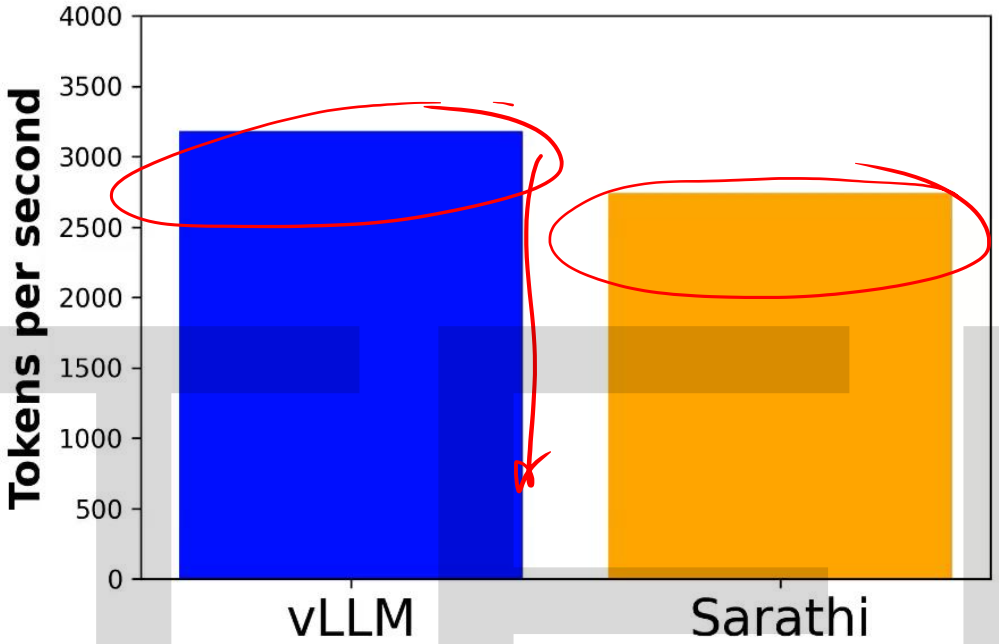
Setup

ShareGPT4 trace on on A100 GPUs with strict (S) and relaxed (R) latency SLOs



5-6x higher capacity 🎊






```
[18:31:13][distributed_c10d.py:476:_store_based_barrier][INFO] Rank 0: Completed sto
re-based barrier for key:store_based_barrier_key:4 with 1 nodes.
[18:31:13][distributed_c10d.py:442:_store_based_barrier][INFO] Added key: store_base
d_barrier_key:5 to store for rank: 0
[18:31:13][distributed_c10d.py:476:_store_based_barrier][INFO] Rank 0: Completed sto
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[18:31:13][distributed_c10d.py:442:_store_based_barrier][INFO] Added key: store_base
d_barrier_key:7 to store for rank: 0
[18:31:13][distributed_c10d.py:476:_store_based_barrier][INFO] Rank 0: Completed sto
re-based barrier for key:store_based_barrier_key:7 with 1 nodes.
INFO 11-10 18:31:21 llm_engine.py:217] # GPU blocks: 2713, # CPU blocks: 1365
Processed requests:   0%|                                          | 0/96 [00:00<?, ?it/s]
```

```
[18:01:18][distributed_c10d.py:476:_store_based_barrier][INFO] Rank 0: Completed st
ore-based barrier for key:store_based_barrier_key:4 with 1 nodes.
[18:01:18][distributed_c10d.py:442:_store_based_barrier][INFO] Added key: store_bas
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[18:01:18][distributed_c10d.py:476:_store_based_barrier][INFO] Rank 0: Completed st
ore-based barrier for key:store_based_barrier_key:7 with 1 nodes.
INFO 11-10 18:01:26 llm_engine.py:217] # GPU blocks: 2749, # CPU blocks: 1365
Processed requests:   0%|                                          | 0/96 [00:00<?, ?it/s]
```

🗒️ Summary

😨 **Problem:** State-of-the-art systems sacrifice decode latency to achieve higher throughput

💡 **Key Insight** - Low arithmetic intensity of decodes allows for adding compute intensive prefills with negligible decode latency cost

🧮 **Key Results** - We achieve optimality in both latency and throughput simultaneously leading up to 6x higher capacity under SLO constraints

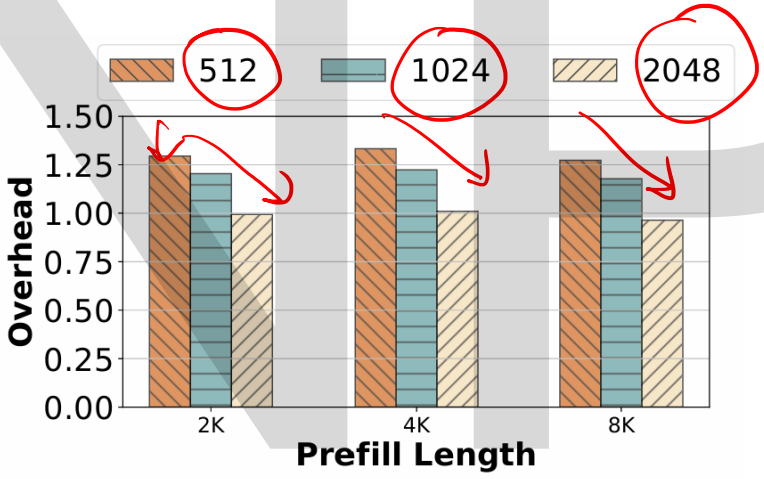
🎉 **Industry Adoption** - Available in all major serving frameworks and more.

vLLM CentML anyscale

Paper GitHub Use in vLLM

Is this a perfect solution?

- **Balancing low latency and high prefill throughput is hard**

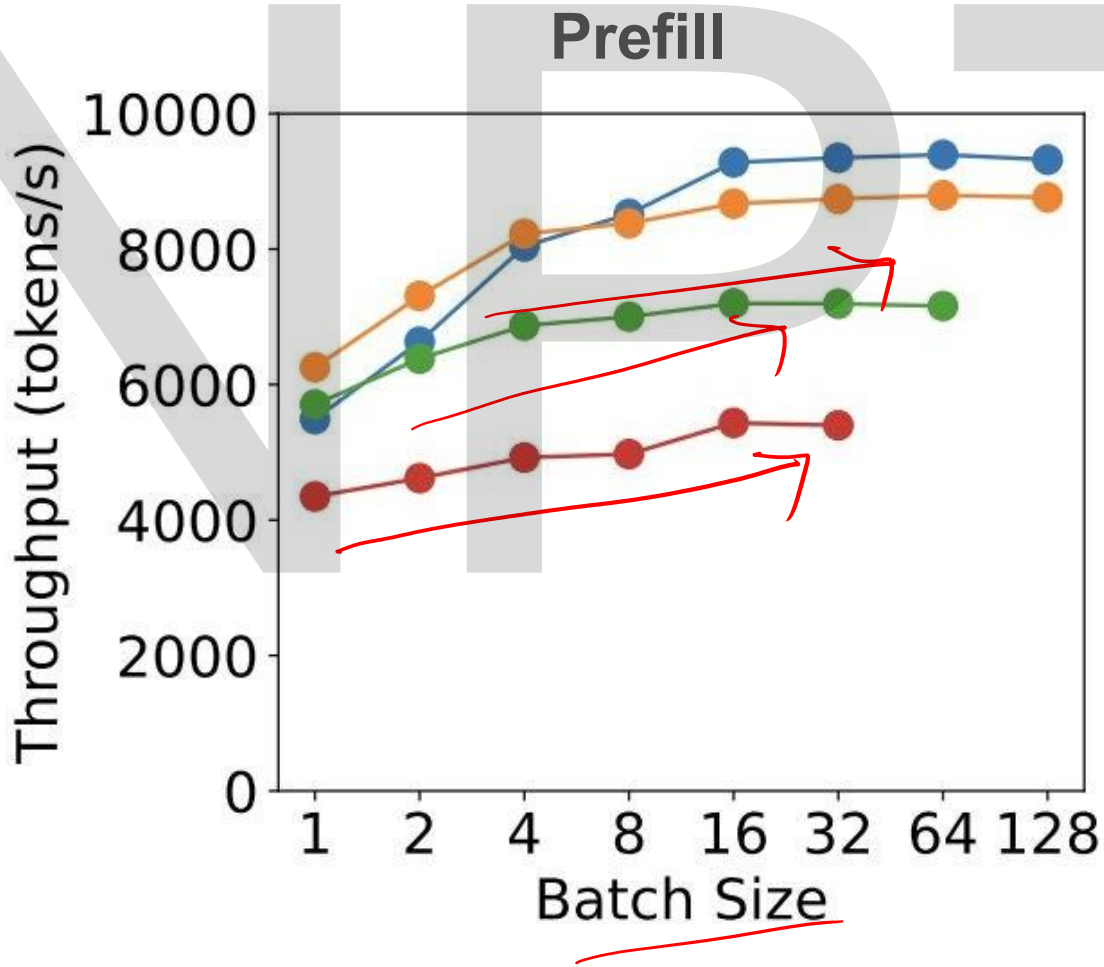


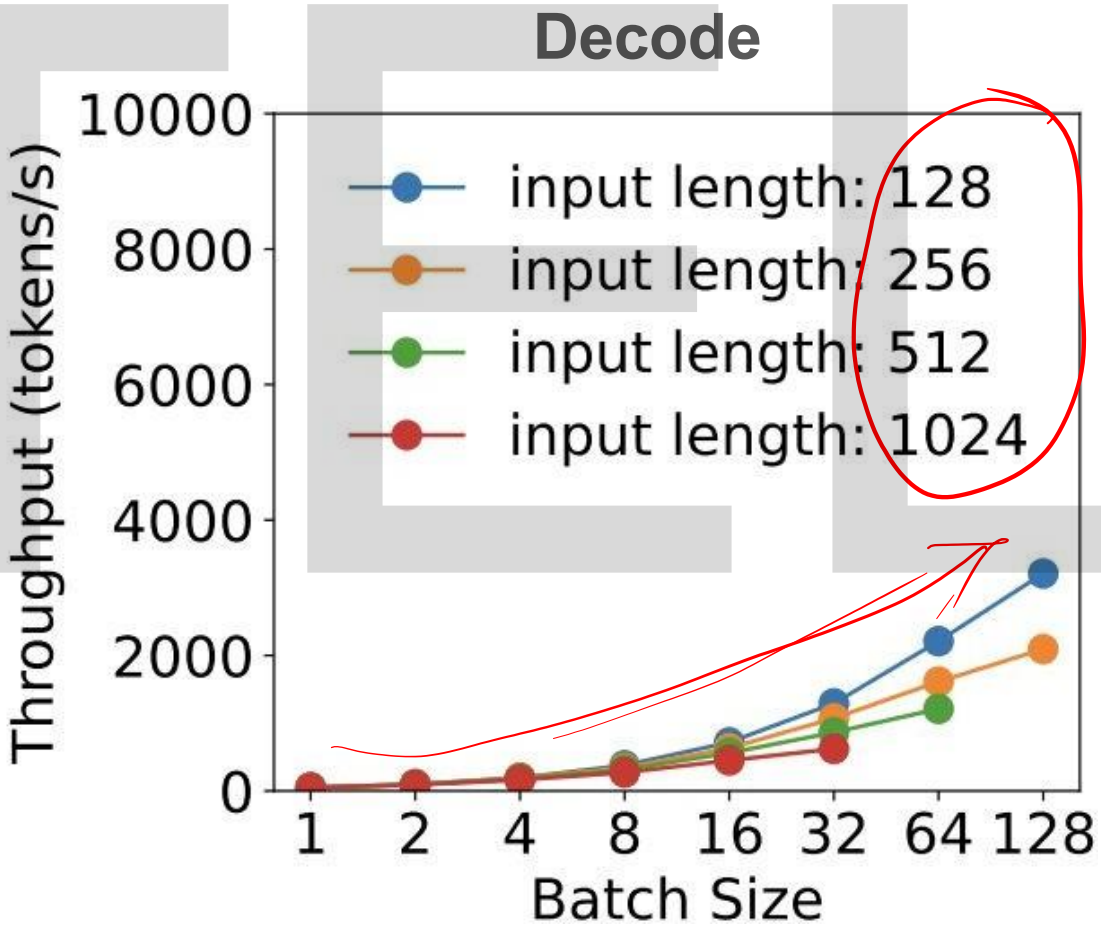
Overhead of chunking increases as chunk size reduces

We need smaller chunks for lower TBT

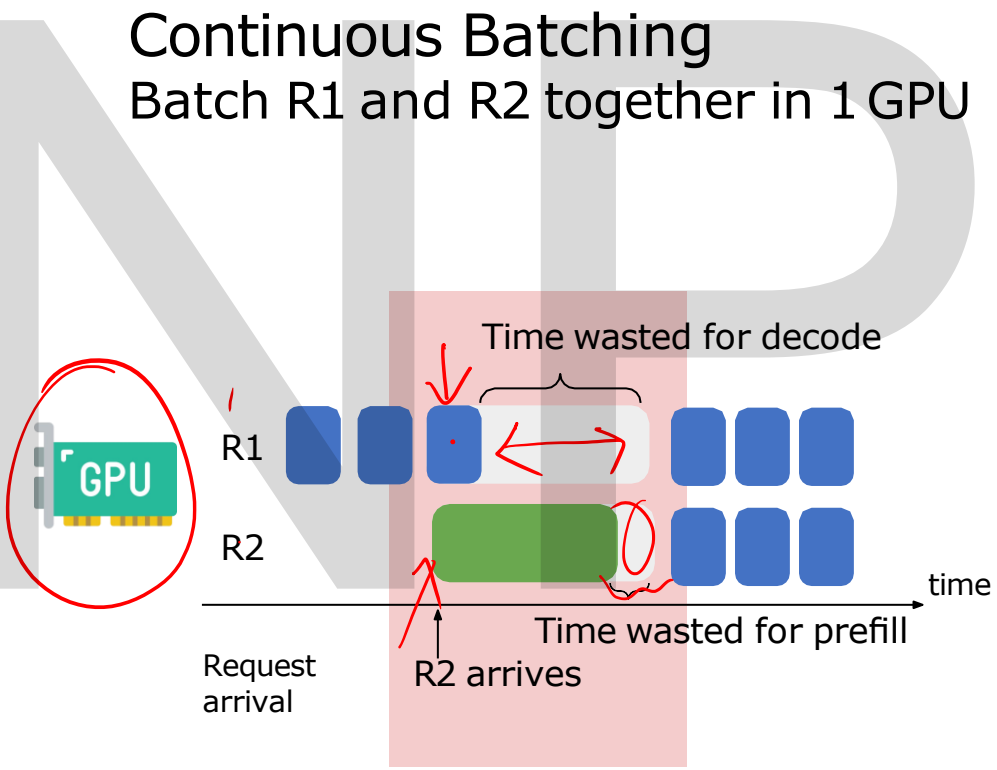
Smaller chunks => lower prefill throughput

Prefill and Decode have Distinct Characteristics



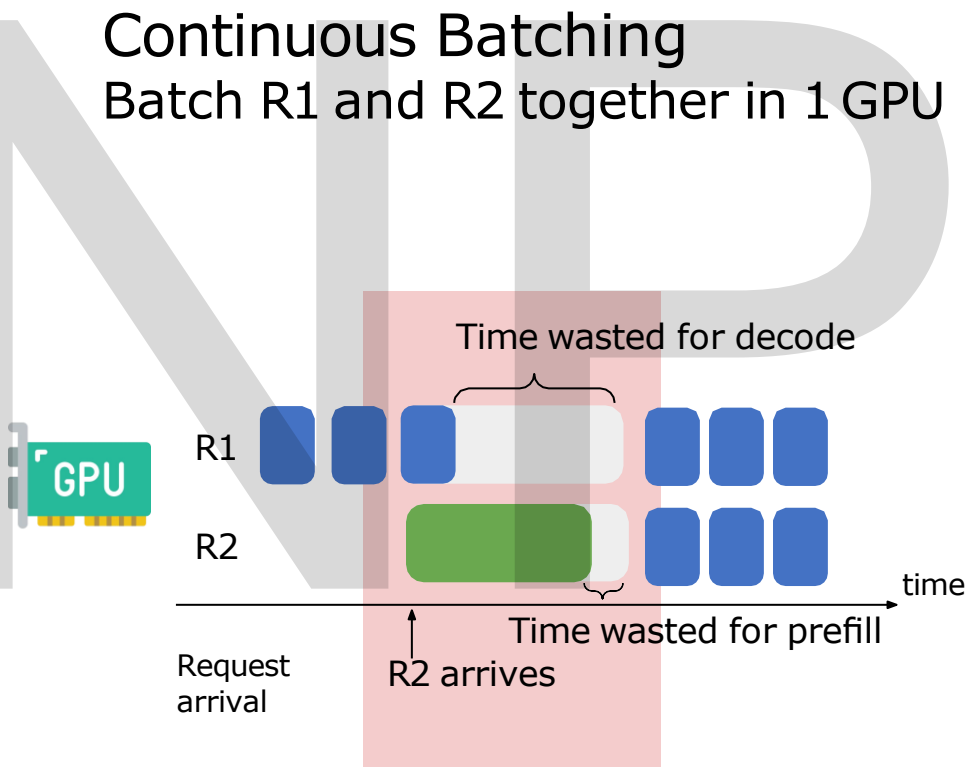


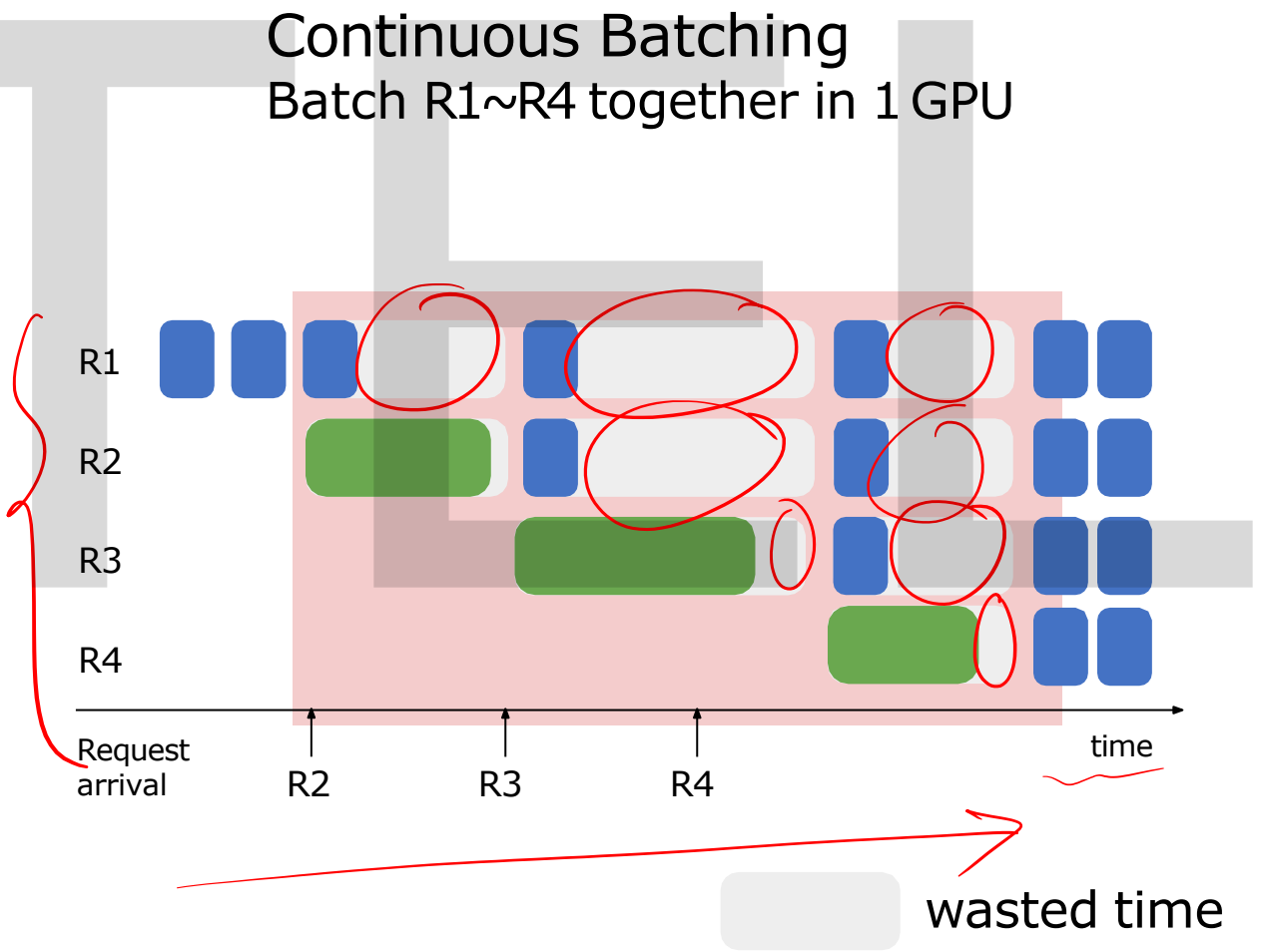
Continuous Batching Causes Interference



wasted time

Continuous Batching Causes Interference







THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)